

RZHEV, Russian Federation

WMO: 264980

Lat: 56.267N

Lon: 34.317E

Elev: 643

StdP: 14.36

Time Zone: 3.00 (E03)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-10.2	-3.5	-14.1	2.6	-8.9	-8.3	3.6	-2.4	15.0	30.3	12.1	30.7	1.7	90	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	80.6	66.9	77.4	65.3	74.6	63.5	69.1	77.4	66.9	74.8	65.1	72.3	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.8	73.7	63.9	90.9	70.9	62.0	85.1	69.0	33.6	76.8	31.8	74.8	30.3	72.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.2	9.7	DB	-18.0	84.8	5.3	4.0	-21.8	87.7	-24.9	90.0	-27.9	92.3	-31.7	95.2	(4)
(5)			WB	-18.2	73.0	5.3	2.5	-22.0	74.7	-25.1	76.2	-28.0	77.6	-31.9	79.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	21.2	21.6	29.9	42.1	53.1	60.4	62.8	60.4	50.3	40.1	28.8	21.0		
	DBStd	17.73	10.35	11.93	7.47	8.38	7.42	6.48	5.42	6.04	7.68	8.11	9.71	10.96		
	HDD50	4568	892	795	622	265	51	2	0	2	87	315	637	900		
	HDD65	8835	1357	1215	1087	687	377	171	105	169	445	770	1087	1365		
	CDD50	1314	0	0	0	28	146	313	399	323	95	10	0	0		
	CDD65	105	0	0	0	0	7	33	38	25	2	0	0	0		
	CDH74	746	0	0	0	1	50	195	304	175	21	0	0	0		
	CDH80	117	0	0	0	0	7	26	60	21	3	0	0	0		
(14) Wind	WSAvg	4.0	5.0	4.9	4.7	3.9	4.1	3.5	3.4	3.1	3.5	4.1	4.0	4.3		
(15) Precipitation	PrecAvg	24.6	1.4	1.2	1.2	1.4	2.3	3.1	3.2	2.8	2.4	1.9	1.9	1.9		
	PrecMax	30.6	3.3	3.3	3.0	3.1	4.4	7.2	6.0	6.9	6.2	5.7	4.5	4.0		
	PrecMin	17.6	0.1	0.2	0.3	0.2	0.6	0.2	0.5	0.4	0.4	0.0	0.0	0.5		
	PrecStd	3.3	0.7	0.7	0.7	0.8	1.1	1.7	1.5	1.7	1.4	1.1	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	42.9	50.0	71.8	80.7	83.4	85.8	82.7	76.4	62.7	49.0	38.3		
		MCWB	36.9	39.5	42.7	54.6	64.4	70.1	68.7	70.2	63.7	54.5	47.1	37.5		
	2%	DB	36.3	40.1	44.5	65.9	73.8	79.2	80.9	78.3	69.8	57.7	45.7	36.0		
		MCWB	34.5	37.9	38.6	52.4	59.2	66.3	67.6	65.7	60.7	52.3	44.2	34.9		
	5%	DB	34.9	37.6	41.4	61.1	70.4	76.0	76.9	75.4	64.5	53.7	42.8	34.8		
		MCWB	33.5	35.5	38.0	49.1	57.9	65.2	65.0	64.2	58.2	50.0	41.7	33.6		
	10%	DB	33.7	35.7	39.2	56.8	66.2	72.7	73.8	72.3	60.6	51.2	40.5	33.2		
		MCWB	32.7	33.8	35.9	47.7	55.8	63.2	63.9	62.8	56.5	48.1	39.5	32.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	40.4	44.5	55.5	65.5	71.2	71.7	71.7	65.5	56.4	47.3	37.8		
		MCDB	38.8	41.7	49.3	68.6	77.9	80.7	83.0	77.4	73.9	59.9	48.6	38.1		
	2%	WB	34.9	37.8	41.1	53.0	62.3	68.4	69.0	67.7	61.6	53.1	44.6	35.4		
		MCDB	35.9	39.7	43.2	63.1	71.3	76.5	77.5	75.5	66.4	55.8	45.7	35.9		
	5%	WB	33.8	36.1	38.4	50.7	59.3	66.4	66.8	65.3	58.9	51.0	41.7	34.0		
		MCDB	34.7	37.6	40.7	59.2	67.1	74.1	74.6	72.7	63.4	53.7	42.5	34.8		
	10%	WB	32.7	34.1	36.3	48.1	56.8	64.4	64.7	63.5	56.7	48.4	39.7	32.2		
		MCDB	33.7	35.3	38.6	55.5	64.5	70.6	71.7	70.6	60.8	50.8	40.6	33.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.3	11.0	14.9	16.4	15.9	15.8	15.9	12.4	10.3	7.2	7.2		
		MCDBR	7.4	7.3	11.8	23.3	22.4	18.9	21.8	22.4	18.2	15.0	6.9	5.4		
		MCWBR	7.2	6.5	8.0	11.5	10.6	8.6	10.5	11.1	10.0	10.0	6.3	5.3		
	5% WB	MCDBR	7.1	7.4	9.0	20.6	18.7	16.8	19.0	19.2	16.4	12.5	6.8	5.1		
		MCWBR	7.0	6.8	7.4	11.1	10.0	9.0	10.9	10.6	9.2	9.2	6.4	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.317	0.337	0.396	0.387	0.368	0.400	0.410	0.364	0.343	0.316	0.277		
	taud		2.133	2.096	2.161	2.196	2.328	2.401	2.327	2.291	2.393	2.388	2.248	2.137		
	Ebn at Noon		189	228	258	259	270	277	265	252	249	223	176	158		
	Edh at Noon		18	29	35	39	37	35	36	36	28	22	17	14		
(44) All-Sky Solar Radiation	RadAvg		171	434	912	1262	1655	1739	1698	1348	872	398	146	96		
	RadStd		34	63	101	120	142	167	179	145	110	75	27	28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+1.24	N/A	N/A	N/A	N/A	N/A	-283	-410	+175	N/A

Nomenclature: See separate page